

The Dark Side of Love: Prediction of Digital Intimate Partner Violence and Associated Factors Among University Students Using Machine Learning

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Abstract

This study aimed to identify key risk factors and predict digital intimate partner violence (DIPV) exposure and perpetration among university students using machine learning (ML) algorithms. A cross-sectional online survey was conducted with 1,764 university students (age range = 18–41 years, $M = 20.8$; 87.2% female, 12.8% male) selected through snowball sampling from a large public university in Türkiye. The survey included sociodemographic, lifestyle, and relationship variables, along with the Digital Intimate Partner Violence Scale. Six ML models were used: Logistic Regression (LR), XGBoost, Gradient Boosting (GB), Random Forest (RF), LightGBM, and Support Vector Machines (SVM). Model performance was evaluated using accuracy, precision, recall, F1 score, and receiver operating characteristics-area under the curve (ROC-AUC). XGBoost achieved the highest performance ($AUC = 0.996$), followed closely by RF and LightGBM ($AUC = 0.995$). LR and GB also performed well

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(AUC=0.992), while SVM had slightly lower performance (AUC=0.989). SHapley Additive exPlanations analysis revealed that domestic violence history, urban residence, father's low education, short relationship duration, and frequent digital communication were risk factors. High income perception and non-smoking reduced DIPV risk. ML models, particularly XGBoost, effectively predict DIPV. Socioeconomic and psychosocial factors should be targeted in prevention efforts, alongside digital literacy and support services.

Keywords

undergraduate, intimate partner violence, machine learning, prediction models, SHAP analysis

Introduction

Intimate partner violence refers to any behavior of an intimate partner or former partner that causes physical, sexual, or psychological harm to the individual, such as physical aggression, sexual coercion, psychological pressure, abuse, and control of behavior (Torp Løkkeberg et al., 2024; WHO, 2021a, 2021b). Dating violence is a subtype of intimate partner violence. Examining the international literature reveals a similar application of the concepts of dating violence and intimate partner violence. This study uses the terms "intimate partner" and "dating" interchangeably.

Digital dating violence is defined as "a broad pattern of repeated abuse (controlling and coercive behaviors) committed by a person (perpetrator) against a current or former intimate partner (victim) through the use of technology, regardless of geographical boundaries" (Novitzky et al., 2023). This term is also known as technology-facilitated violence (British Columbia Society of Transition Houses, 2021). Digital dating violence is an umbrella term for behaviors that involve controlling, pressuring, or threatening a dating partner using digital technologies. Digital technologies include social media, online games, multimedia, smartphones, or the internet (Monteiro et al., 2023).

The term digital dating violence can include relationship abuse (sharing embarrassing photos via electronic means, unwanted calls, cyberstalking); control (surveillance, monitoring, spyware applications, hacking social media accounts); harassment (revenge porn, reputation damage); attempts to isolate the victim (tampering with, destroying, hiding assistive technology devices); and existential threat (making hostile threats, killing; Gilbar et al., 2023; Novitzky et al., 2023; Rodríguez-deArriba et al., 2021).

The concept of digital intimate partner violence (DIPV) has been studied for the last 10 years as a form of violence that can cause psychological harm to its victims. Although the concept of violence is a much-discussed topic, how this phenomenon differs between face-to-face violence and digital violence has been little researched (Gilbar et al., 2023). Although DIPV is an inclusive concept that refers to acts of violence committed on all digital platforms, the fact that these acts are carried out over social media due to the intensive use of social media today increases the importance of the issue for every individual. DIPV is reported to affect millions of people annually (Glass et al., 2024). Therefore, it is an important social problem. According to the Digital Violence in Turkey Research conducted by the Social Information and Communication Association in 2021, social media platforms are the most common platforms where acts of digital violence are committed (Instagram 53%, Facebook 35%, Twitter 19%; Sener & Abınık, 2021). Considering the intensity of social media use in Turkey, the increasing use of digital violence methods, and the negative effects created by these methods, a close examination of the issue is required.

Experiencing digital violence causes serious short- and long-term problems for individuals. These include physical (physical injuries, fractures, bruises, disabilities), psychological and behavioral (decreased self-esteem, insecurity, fear, anxiety, emotional distress, feelings of shame and guilt, phobias, post-traumatic stress, self-harm, sleep problems, substance abuse), sexual and reproductive problems (unsafe abortions and unwanted pregnancies, sexually transmitted infections), and even fatal consequences such as homicide, suicide, and maternal mortality (Maftai & Dănilă, 2023; Novitzky et al., 2023; Spencer et al., 2021; Torp Løkkeberg et al., 2024). In a study, it was found that as a result of violence, approximately one-third of individuals experienced poor mental health in the last 30 days, and half of them felt constant feelings of sadness and hopelessness (Glass et al., 2024). Furthermore, Glass et al. (2024) reported that one-fifth of the individuals had contemplated suicide, indicating a significant risk. Mental health comorbidities such as anxiety, post-traumatic stress disorder, and depression are three to five times more common in these individuals than in the general population (Emezue et al., 2022). Therefore, it is critical to draw attention to the importance of this issue.

Artificial intelligence and machine learning (ML) are applications of artificial intelligence that can learn and perceive with data. ML has been successfully used in many situations where it is difficult or impossible to use traditional algorithms to accomplish any task. The most important advantage of ML is its ability to make consistent and high-performance predictions using complex and non-linear relationships between features. Artificial intelligence is very successful in making predictions based on strong statistical

analysis applied to existing data (Novitzky et al., 2023). Therefore, it is important to develop systems that can help in the investigation of this phenomenon. DIPV is a public health problem that threatens both the mental and physical health of the individual. In the age of rapidly advancing technology, there is a need to base new studies on evidence-based data in solving the problems of violence as the ML field advances (Hui et al., 2023). With the unprecedented increase in data available on the Internet and electronic record systems, utilizing ML to detect ambiguous changes and predict likelihood from digital text data is a promising area of research. However, there is a paucity of research discussing and reviewing ML applications in violence research (Hui et al., 2023). In contrast to conventional regression approaches, machine learning methods are capable of handling large-scale, complex, and multidimensional datasets while capturing non-linear interactions between variables. This ability provides greater predictive accuracy and deeper insights into hidden risk factors of DIPV. Additionally, the use of explainability tools such as SHapley Additive exPlanations (SHAP) enhances transparency in model outputs, which strengthens their practical utility in public health and violence prevention research.

The study is also conceptually informed by Cyber-Routine Activities Theory (CRAT), which emphasizes that routine behaviors in digital environments—such as frequent online interactions or prolonged social media participation—can increase opportunities for digital victimization by presenting motivated offenders with suitable targets in the absence of effective guardianship in cyberspace (Vakhitova, 2024). Previous research grounded in similar frameworks (Choi & Lee, 2017; Van Ouytsel et al., 2018) supports the relevance of CRAT-based explanations in explaining cyber-interpersonal violence dynamics.

This study was conducted in Türkiye, where cultural norms regarding gender roles, romantic relationships, and digital communication may influence both the occurrence and reporting of DIPV. For example, qualitative research with university students in Türkiye revealed that motivations such as anger, jealousy, and fear of loss frequently underpin digital dating violence behaviors (Bacıoğlu & Akçay, 2024). Surveys among Turkish nursing students also indicate that cyber-dating abuse is recognized and experienced within healthcare-related cohorts, underlining the importance of digital literacy and awareness in prevention efforts (Er et al., 2022). Additionally, attachment styles—an important psychosocial factor—have been shown to predict cyber-dating violence behaviors in Turkish university populations (Erdem et al., 2022). These findings suggest that digital IPV in Türkiye is contextually shaped and warrants culturally informed analysis. In addition, machine learning methods are particularly suitable for analyzing DIPV, as they can capture non-linear and

high-dimensional relationships among demographic, psychosocial, and behavioral factors, while also providing predictive accuracy and interpretability. These methodological strengths are critical for uncovering hidden patterns in sensitive and complex phenomena such as digital partner violence.

In the context of DIPV, risk factors such as digital communication patterns, psychosocial characteristics, prior exposure to violence, and socioeconomic status may not operate in isolation. Their effects often interact in non-linear ways—certain combinations of risk factors may amplify vulnerability that would remain undetected under linear regression models. For example, extensive digital contact may escalate to DIPV chiefly when accompanied by prior victimization or limited offline support. Machine learning techniques, including XGBoost and Random Forest, excel at capturing such non-linear interactions and hidden dependencies, enabling more accurate prediction and deeper insights into the interplay of complex risk factors in sensitive phenomena such as digital violence (Cruz-Mendoza et al., 2025; Hui et al., 2023). Unlike traditional statistical approaches frequently used in IPV research, such as logistic regression, multilevel modeling, or structural equation modeling, ML methods do not rely on strict assumptions of linearity or normality and can handle high-dimensional datasets with multicollinearity. This allows ML to uncover complex, non-linear interactions and improve predictive performance, thereby complementing rather than replacing conceptually driven methods. Prior research has also demonstrated the utility of ML in violence research. For example, ML models have been used to classify high-risk victims and predict episodes of IPV (Hui et al., 2023), to enhance prevention strategies by detecting complex patterns across diverse contexts (Cruz-Mendoza et al., 2025), and to analyze cyber dating abuse dynamics among youth populations (Erbiçer et al., 2025). These applications provide critical evidence of ML's added value for understanding and predicting complex social problems such as DIPV.

The aim of this study is to predict DIPV using ML models.

Research questions:

RQ1: What is the prevalence of DIPV among university students in the study sample?

RQ2: Which demographic (e.g., age, gender, family income), psychosocial (e.g., prior exposure to violence, psychiatric history), relational (e.g., relationship status, duration, family response), and contextual (e.g., region, digital communication patterns) factors are most strongly associated with DIPV among university students, as identified by ML models and conceptually informed by CRAT?

RQ3: Which ML algorithm demonstrates the highest predictive performance in classifying DIPV cases?

Method

Type of Research

This study is a cross-sectional study that includes a data mining-based analysis that aims to predict DIPV using ML models.

Population and Sample

The population of the study consisted of students of a University ($n=46,845$). The study was conducted at a large public university in western Türkiye with more than 46,000 enrolled students across diverse faculties. As a public institution, it attracts students from different socioeconomic and geographical backgrounds, providing a heterogeneous sample; however, as a single-site university, its student population may not fully represent the diversity of Türkiye's entire higher education system. The snowball sampling method, which is one of the non-probability sampling methods, was used in the study. However, with a "margin of error of 3%" and " p and q 0.5," the estimated sample size was determined as 1,044 if the population was over 20,000. The research was completed with 1,764 participants.

Data Collection Method

Data were collected online via Google Forms between January 1 and March 1, 2025. 5 min was allocated for each questionnaire.

Data Collection Tools

A questionnaire form and the Digital Intimate Partner Violence Scale (DIPVS) were used for data collection. Participants were initially recruited through announcements on university mailing lists, student social media groups, and class communication platforms. After completing the survey, participants were encouraged to share the survey link with peers in their networks, thereby enabling the snowball sampling process. All participants were informed that their participation was voluntary, they could withdraw at any time without consequence, and no identifying information (such as names, student IDs, or IP addresses) was collected. Data were stored securely and analyzed in aggregate form only, ensuring full anonymity and confidentiality.

Questionnaire Form: This form was prepared in line with the literature (Gecim & Ardic, 2023; Sarıcam & Sinsoysal, 2024; Sarısın, 2022; Seçgin & Selçuk, 2023; M. E. Yıldız, 2022; F. Yıldız & Ozarslan, 2021) consisting of a total of 29 questions including age, gender, mother and father's education level, income and expenditure perception of herself and her family, smoking and alcohol use status, presence of intimate partner, duration of relationship, thinking about marriage with the person being dated, frequency of meeting with the intimate partner, duration of communicating with the partner in digital environment during the day, thinking about being exposed to violence in dating relationship, thinking about being subjected to violence in dating relationship, being exposed to domestic violence.

Digital Intimate Partner Violence Scale: The Turkish validity and reliability of the scale developed by López-Cepero et al., 2021 were conducted by Sarıcam and Sinsoysal (2024). The 12-item DIPVS is divided into two sub-dimensions: destruction-oriented cyber abuse (1, 2, 4, 5, 8, 10, 11) and control-oriented cyber abuse (3, 7, 9, 12). The scale is a 5-point Likert scale (1: *Never*, 2: *Rarely*, 3: *Sometimes*, 4: *Usually*, 5: *Always*), and the lowest score is 11 and the highest score is 55. The scale's Cronbach alpha coefficient is .958. In this study, the Cronbach alpha coefficient was .987. This scale was preferred because it has been validated for use in the Turkish context, ensuring cultural and linguistic appropriateness, and because it demonstrates high reliability (Cronbach α = .987 in this study), making it a robust tool for measuring DIPV among Turkish university students. The selection of the DIPVS was conceptually grounded in prior literature on digital partner violence and cyber-abuse, as it captures both control-oriented and exploitation-oriented forms of digital IPV. Its strong psychometric properties in the Turkish context (Cronbach's α = .987 in this study) provide robust construct validity, ensuring that the measure appropriately reflects the theoretical domains under investigation.

Data Analytics and Machine Learning Models

In this study, ML models were used to identify factors associated with DIPV and to predict individuals' exposure to violence. The data analysis process consists of preprocessing, modeling, and performance evaluation.

Data Preprocessing

Prior to data analysis, data cleaning and transformation procedures were performed to increase the accuracy of the model and to prevent erroneous results. The data set was examined for missing data, and no missing data was

observed. The Z-Score method was used to detect outliers, but no extreme outliers were observed. The DIPV score was calculated as a continuous variable and then transformed into categorical classes according to certain threshold values. Categorical variables were transformed with label encoding. Numerical variables were scaled with the Min-Max Scaling method. To maintain proportional representation of classes, stratified sampling was used in the train-test split, ensuring that the prevalence of DIPV-positive cases (21.3%) was preserved across both training and testing sets.

Data Analysis

A statistical evaluation was performed with descriptive statistics, an independent sample *t*-test, and Pearson correlation tests in the SPSS 22.0 package program. Significance was evaluated at the $p < .05$ level. In addition, the analyses were conducted in the Google Co laboratory environment using the Python 3.0 programming language, and the following libraries were used:

- **Scikit-learn:** for ML models.
- **SHAP:** is used to describe the outputs of the model.
- **Matplotlib and Seaborn:** for visualization of data and model outputs.

Prior research also illustrates ML's added value in IPV contexts. Shashidhara et al. (2024) employed LASSO, Random Forest, and Gradient-Boosted Tree models to develop a 15-question tool for identifying high-risk IPV cases with high recall—insights unlikely to emerge via linear modeling. Similarly, Chen et al. (2023) used ML to estimate IPV incidence rates in China, effectively handling missing and unbalanced data. These applications affirm that DIPV's multifactorial predictors and interaction effects may be better captured by ML approaches, reinforcing their appropriateness in our study.

Machine Learning Models

Six machine learning models were used in this study: Logistic Regression (LR), Extreme Gradient Boosting (XGBoost), Gradient Boosting (GB), Random Forest (RF), Light Gradient Boosting Machine (LightGBM), and Support Vector Machine (SVM). These models were used to find the factors that are linked to DIPV and to predict how often people will be exposed to violence.

The reason for choosing these models was to compare the predictive power of different algorithms, the importance of variables, and determine which method provided the best performance. The dataset was initially

divided into 80% training and 20% testing. To minimize overfitting and provide more robust performance estimates, we additionally applied stratified 10-fold cross-validation repeated three times. Model performance was reported as mean $\pm$ standard deviation for all evaluation metrics (accuracy, precision, recall, F1 score, and area under the curve (AUC)). In the context of DIPV, such methods allow us to model the complex and non-linear relationships that arise from the interaction of psychosocial, demographic, and behavioral variables, providing a more nuanced understanding of risk patterns.

Model Evaluation

In this study, five different metrics are used to evaluate the performance of six different ML models. The performance of each model is measured by these metrics, and the best-performing model is identified.

Accuracy: Accuracy indicates the rate at which all predictions of the model are correct. TP (true positive): Classes correctly predicted as positive, TN (true negative): Classes correctly predicted as negative, FP (false positive): False positive predictions (the model predicts a positive state when it is actually negative), FN (false negative): False negative predictions (the model predicts a negative state when it is truly positive).

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision: Precision shows how accurate the model's positive predictions are. A high Precision indicates that the model minimizes false positives (FP).

$$\text{Precision} = TP / (TP + FP)$$

Recall—Sensitivity: Recall shows how accurately the model can capture true positives (TP). A high recall indicates that the model minimizes false negatives (FN). In DIPV prediction, high recall is important to avoid missing at-risk individuals.

$$\text{Recall} = TP / (TP + FN)$$

F1 Score: F1 Score: The F1 score is the harmonic average of Precision and Recall. It provides a more balanced assessment.

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Table 1. Parameters Optimized in Hyperparameter Optimization.

Parameter	Explanation	Optimum values
n_estimators	Number of trees	100–500
learning_rate	Learning rate	0.01–0.1
max_depth	Maximum depth	3–10
min_samples_split	Minimum number of splitting samples	2–10
min_samples_leaf	Minimum number of leaves	1–5
subsample	Sampling rate	0.7–1.0
colsample_bytree	Feature selection rate	0.6–1.0

Model Comparisons with ROC-AUC Curve: The receiver operating characteristic (ROC) curve and the area under the receiver operating characteristic (AUC-ROC) are two of the most critical metrics that indicate the overall performance of the models. The closer the AUC value is to 1, the more successful the model is. A model with an AUC value close to 0.5 is randomly predicting.

Hyperparameter Tuning

Hyperparameter optimization is a critical step to improve the success of ML models. The model does not learn the hyperparameters during training, which means we must manually optimize them. Optimal hyperparameter selection reduces the risk of overfitting (overlearning) while increasing the accuracy of the model. This study performs hyperparameter tuning for the optimization of three gradient boosting-based models (XGBoost, GB, and LightGBM). Gradient boosting is a boosting method that focuses on the errors of the model at each iteration to minimize the error. Therefore, choosing the right hyperparameter directly affects the performance of the model.

This study used two different methods for hyperparameter tuning: grid search and random search.

We first used random search to determine a general range of hyperparameters, and then we used grid search to optimize the best combination. In the hyperparameter optimization process, the following parameters were optimized for the models (Table 1):

SHAP Analysis

SHAP analysis was applied to provide transparency of the model outputs and to understand which features contribute more to the target variables. SHAP

analysis is done to explain the output of an ML model and to understand which features have how much impact on the predictions on the target variable. It shows which features are more influential in the model and whether these effects are positive (increasing the prediction) or negative (decreasing it). Understanding why the model makes a particular prediction increases the reliability of model outputs.

- **Global Analysis:** The order of importance of the features is examined over the entire dataset (SHAP Summary Plot).
- **Local Analysis:** We analyzed which features contribute to the model prediction of individual observations (SHAP Force Plot).

Ethical Considerations

Permission was obtained from the Ethics Committee for the conduct of the study (Decision Number: 2024-YÖNP-0226/05/79). Written informed consent was also obtained from the participants. There was an informed consent form at the beginning of the questionnaire, and there was a check box in the questionnaire indicating that it was completed voluntarily. Participants were informed that participation in the study was voluntary and that the data collected would only be used for scientific purposes.

Results

The frequency distribution of the descriptive information of the students who participated in the study is given in Table 2.

The mean age of the students was 20.82 ± 2.35 years; the majority of them were female (87.2%) and single (98.6%); the most common education level of their mothers was primary school (37.1%), and their fathers were high school graduates (32.1%). 52.9% of the students reported having an intimate partner, while 47.3% reported having no relationship. Of those who were dating, 40.8% were planning to get married, and the most common duration of digital communication was 3 to 4 hr (33.3%). The rate of those who have been subjected to intimate partner violence is 21.3%, while the rate of those who have been subjected to partner violence is 10.7%. The rate of those who have been subjected to domestic violence is 17.7%, and the rate of those who stated that they have a violent family member is 22.4% (Table 2). This prevalence (21.3% positive vs. 78.7% negative) was preserved in both training and test subsets through stratified sampling.

Table 2. Frequency Distribution of Descriptive Information of the Participants ($n = 1,764$).

Variables	$X \pm SD$	
Age	20.82 $\pm$ 2.35 (min:18 max:41)	
	<i>n</i>	%
Gender		
Female	1,539	87.2
Male	225	12.8
Class		
First grade	594	33.7
Second grade	540	30.6
Third grade	288	16.3
Fourth grade	342	19.4
Marital status		
Married	24	1.4
Single	1,740	98.6
Family type		
Nuclear family	1,479	83.8
Extended family	216	12.2
Single-parent family	69	3.9
Mother's education level		
Illiterate	81	4.6
Literate	78	4.4
Primary school graduate	654	37.1
Secondary school graduate	321	18.2
High school graduate	417	23.6
University graduate	189	10.7
Postgraduate	24	1.4
Father's education level		
Illiterate	18	1.0
Literate	33	1.9
Primary school graduate	462	26.2
Secondary school graduate	369	20.9
High school graduate	567	32.1
University graduate	282	16.0
Postgraduate	33	1.9
Perception of own income status		
Income is less than expense	750	42.5
Income is equal to expense	903	51.2
Income is more than expense	111	6.3
Perception of family income status		
Income is less than expense	375	21.3
Income is equal to expense	1,041	59.0

(continued)

Table 2. (continued)

	<i>n</i>	%
Income is more than expense	348	19.7
The place where most of life is spent		
Metropolitan	810	45.9
Province	396	22.4
District	417	23.6
Village town	141	8.0
Hometown region		
Marmara Region	603	34.2
Aegean Region	204	11.6
Black Sea Region	225	12.8
Mediterranean Region	147	8.3
Central Anatolia Region	231	13.1
Eastern Anatolia Region	231	13.1
Southeast Anatolia Region	123	7.0
Smoking use		
Yes	624	35.4
No	1,140	64.6
Alcohol use		
Yes	594	33.7
No	1,170	66.3
Working status		
Yes	276	15.6
No	1,488	84.4
Presence of a close partner (flirt, lover)		
Yes	930	52.9
No	834	47.1
Duration of the relationship		
No relationship	834	47.3
Less than 1 year	393	22.3
1–3 years	447	25.3
4 years and above	90	5.1
Considering marrying the person you are dating		
Yes	720	40.8
No	594	33.7
Undecided	450	25.5
Family response to the presence of an intimate partner		
I don't let them know because they don't like this kind of relationship	642	36.4
When they know, they find it natural and want to meet	867	49.1
When they know, they don't let me meet	69	3.9

(continued)

Table 2. (continued)

	<i>n</i>	%
They are very happy, and I share my happiness with my family	186	10.5
Frequency of meeting with the intimate partner		
Every day	438	24.8
Every other day	456	25.9
Once a week	411	23.3
Once a month	459	26.0
Time spent communicating digitally with a close partner		
2 hr	567	32.1
3–4 hr	588	33.3
5–6 hr	333	18.9
7 hr and more	276	15.6
Thinking you are experiencing intimate partner violence		
Yes	375	21.3
No	1,233	69.9
Undecided	156	8.8
Thinking you are being violent toward an intimate partner		
Yes	189	10.7
No	1,452	82.3
Undecided	123	7.0
Exposure to domestic violence		
Yes	312	17.7
No	1,452	82.3
Presence of individuals exposed to domestic violence in the family		
Yes	396	22.4
No	1,368	77.6
Presence of diagnosed psychiatric illness		
Yes	150	8.5
No	1,614	91.5
Use of psychiatric medications		
Yes	99	5.6
No	1,665	94.4
Getting psychological support		
Yes	132	7.5
No	1,632	92.5

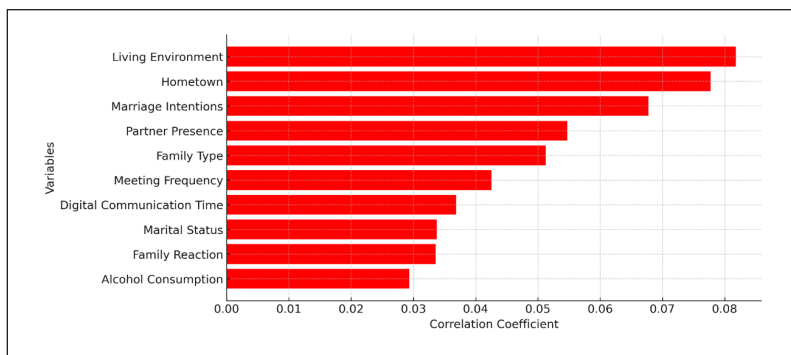
Table 3 presents the participants' scores on the DIPV scale and its sub-dimensions in comparison with the original scale values.

The mean DIPV score of the students was 13.62 ± 7.34 , indicating that the level of DIPV was low. The participants scored 8.53 ± 4.97 on the destruc-

Table 3. Participants' Scores on the DIPV.

Scale and sub-dimensions	Original scale min–max	$X \pm SD$ taken from this study	In this research, min–max
DIPV total score	11–55	13.62 ± 7.34	11–55
Sub-dimension of cyber exploitation focused on destruction	7–35	8.53 ± 4.97	7–35
Control-oriented cyber abuse sub-dimension	4–20	5.09 ± 2.64	4–20

Note. DIPV = digital intimate partner violence.

**Figure 1.** Factors with the highest positive correlation with the DIPV total score.

Note. DIPV = digital intimate partner violence.

tion-oriented cyber abuse sub-dimension and 5.09 ± 2.64 on the control-oriented cyber abuse sub-dimension (Table 3).

Figure 1 displays the variables that have the strongest positive correlation with the DIPV total score. It also shows the strong connections between people's living environment, relationship dynamics, and social environment and digital partner violence.

The largest (but still small) positive correlations were observed with the place where the individual spent most of his/her life ($r \approx .085$) and hometown ($r \approx .08$). Variables such as thinking about marriage ($r \approx .07$), dating ($r \approx .06$), and family type ($r \approx .055$) were also found to be significantly related to digital violence. As the duration of face-to-face contact with the partner ($r \approx .05$) and the duration of digital communication ($r \approx .04$) increased, the DIPV

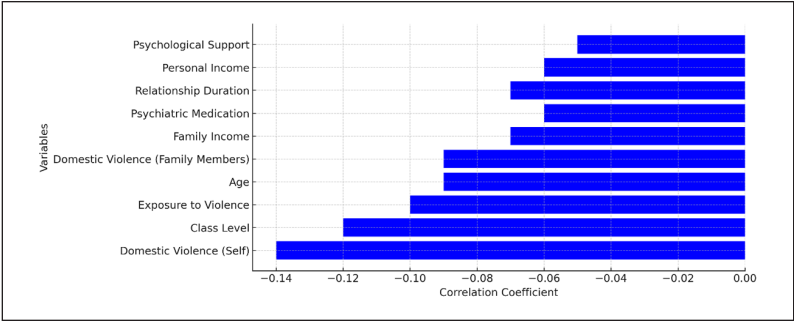


Figure 2. Factors with the highest negative correlation with the DIPV total score. Note. DIPV = digital intimate partner violence.

score also increased. Marital status ($r \approx .035$) and family reaction ($r \approx .03$) also have a significant relationship with digital partner violence. Alcohol use ($r \approx .025$) also showed a positive relationship with the DIPV total score (Figure 1). Figure 2 shows the variables with the highest negative correlation with the DIPV total score.

The largest negative (but small) correlation was observed with exposure to domestic violence ($r \approx -.14$), grade level ($r \approx -.12$), and exposure to violence ($r \approx .10$). As age increases ($r \approx -.09$) and relationship duration increases ($r \approx -.06$), the digital violence score decreases. Family income level ($r \approx -.07$) and having one’s own income ($r \approx -.05$) were also found to be negatively associated with digital partner violence. Receiving psychological support ($r \approx -.04$) and taking psychiatric medication ($r \approx -.06$) were also associated with a decrease in digital partner violence scores (Figure 2). Table 4 shows the comparison of six different ML models used to predict factors associated with DIPV in terms of performance metrics such as accuracy, precision, recall, F1 score, and ROC-AUC.

In Table 4, the LR, XGBoost, and GB models gave very similar results in terms of accuracy (97.4%), precision (97.3%–97.4%), recall (97.4%), and F1 score (97.3%–97.4%). In terms of AUC-ROC value, the XGBoost model (99.6%) achieved the highest value compared to the other models, indicating that it is the model that best discriminates digital violence states. The RF model (96.6% accuracy, 99.5% AUC-ROC) also performed well, but with slightly lower precision (96.7%) and F1 score (96.2%). The LightGBM model performed strongly with an accuracy of 95.7% and an AUC-ROC of 99.5% but lagged behind models such as LR and XGBoost. The SVM model performed significantly worse than the other models, with poor accuracy

Table 4. Comparison of Average Performance Metrics for ML Models (Stratified 10-Fold Cross-Validation).

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
LR	0.974	0.973	0.974	0.973	0.992
XGBoost	0.974	0.973	0.974	0.974	0.996
GB	0.974	0.974	0.974	0.973	0.992
RF	0.966	0.967	0.966	0.962	0.995
LightGBM	0.957	0.956	0.957	0.956	0.995
SVM	0.847	0.718	0.847	0.777	0.989

Note. The performance metrics in Table 4 represent average values obtained from stratified 10-fold cross-validation. ML = machine learning; LR = Logistic Regression; GB = Gradient Boosting; RF = Random Forest; SVM = Support Vector Machines.

(84.7%), precision (71.8%), and F1 score (77.7%). In particular, although the AUC-ROC value is 98.9%, it remains low in other metrics, indicating that SVM cannot learn complex relationships in the dataset well enough.

The very high AUC values observed (>0.99) suggest excellent model performance but also raise the possibility of overfitting. To address this concern, we applied stratified 10-fold cross-validation to reduce bias and provide more reliable estimates of predictive performance. While the results indicate strong discriminative ability, their generalizability to other populations should be interpreted with caution.

The ROC/AUC curves in Figure 3 show that all models are quite successful in predicting the probability of DIPV. The XGBoost model (AUC=0.996) had the highest discrimination power, while the RF (AUC=0.995) and LightGBM (AUC=0.995) models also performed forcefully. The LR (AUC=0.992) and GB (AUC=0.992) models also performed well, while the SVM (AUC=0.989) model performed slightly worse than the others.

Figure 4 shows the graph of SHAP interaction values. This graph visualizes the interactions of gender, age, and class, the three most important variables in predicting the DIPV total score, both on their own and with each other.

Although gender, age, and grade variables alone are determinative in predicting the DIPV score, the interactions of these variables with each other are also important. There is an impression that the contribution to the digital violence score increases for female students at higher grade levels. While the joint effect of age and gender is weaker, the interaction of the grade variable both on its own and with gender is particularly noteworthy (Figure 4).

Figure 5 is the SHAP force plot visualization for the DIPV total score prediction for an individual participant. This plot shows which variables

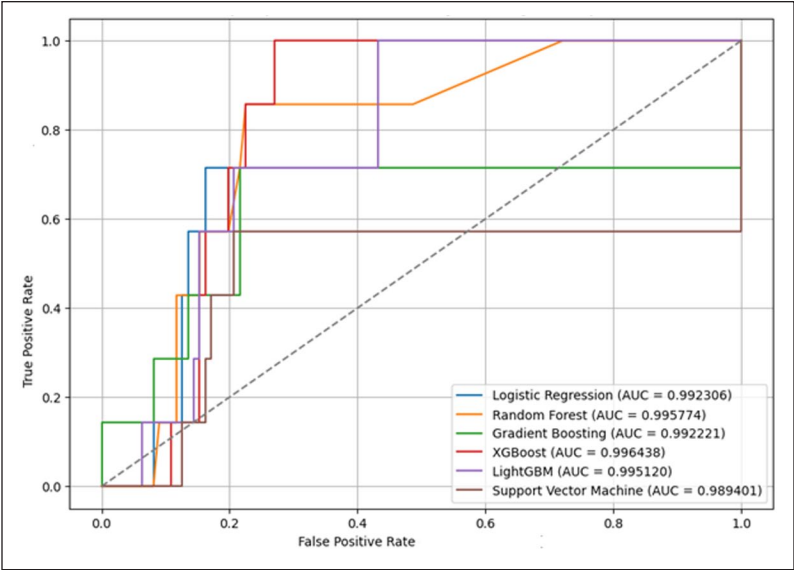


Figure 3. ROC/AUC curves for the models.

contribute to the model’s prediction for that individual, in which direction, and to what extent. The force plot explains in detail how the predicted value (output value) is generated using the SHAP values.

The thick white box in the middle of Figure 5 (2.20): The DIPV total score or risk score value predicted by the model for that individual. Left side (red area): Factors that increase this predictive value. It indicates that the value of the variable contributes to increasing the DIPV score. Right side (blue area): These are factors that decrease the predictive value. It indicates that the variable has an effect on decreasing the digital violence score. The fact that the participant has been exposed to domestic violence increases the DIPV score prediction of the model. Digital partner violence scores increase for those living in metropolitan areas. The fact that the participants’ fathers graduated from primary school was found to be one of the factors that increased the model’s violence prediction. As the age of the participants increases, the digital partner violence score may increase. These findings also align with an ecological framework, which emphasizes that individual-level characteristics (e.g., substance use, attachment patterns) interact with contextual-level influences (e.g., urban residence, regional cultural norms) to shape IPV risk. Such a perspective underscores the multi-layered nature of DIPV. The fact that the

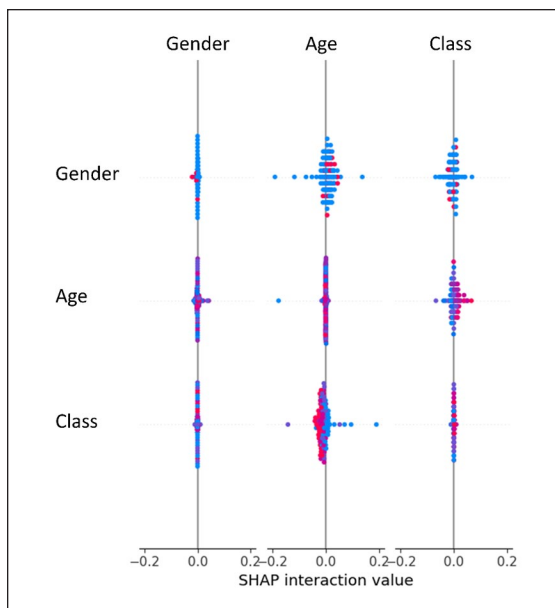


Figure 4. SHAP graph.

Note. SHAP = SHapley Additive exPlanations.

participants were from the Southeastern Anatolia Region led to a lower prediction of the digital partner violence score. The fact that the participants' relationship lasted less than 1 year had a decreasing effect on the digital violence score. The fact that the income status of the participant's family was equivalent to their expenses was considered one of the factors reducing the risk of digital partner violence. The fact that the participant perceived his/her income status as equivalent to his/her expenses was determined as a factor decreasing the digital partner violence score. The fact that the participant did not smoke was found to be a factor that decreased the digital partner violence score. The fact that the participant communicated with his/her partner digitally for less time was evaluated by the model as a risk-increasing factor.

Discussion

In recent years, digital technologies, such as a multitude of digital applications and social networking sites, have provided new opportunities for individuals to initiate, develop, and maintain intimate partner relationships, while facilitating opportunities for individuals to have negative experiences

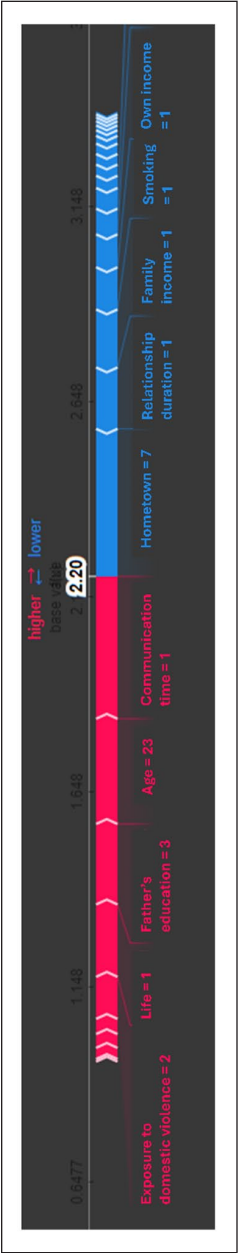


Figure 5. SHAP force plot.
Note. SHAP = SHapley Additive exPlanations.

and perpetrate harm (Henry et al., 2020). As technology advances, new opportunities for partners to exert power and control in their romantic relationships have emerged. International research over the past decade suggests that intimate partner violence is a significant public health problem (Glass et al., 2024). However, regarding the distribution of this phenomenon, the existing literature presents very inconsistent rates of both victimization and perpetration. The main reasons for these different findings stem from the different terminologies, definitions, and times used by each study. Therefore, it is difficult to make direct comparisons or draw valid conclusions between the prevalence rates of each study (Kim & Ferraresso, 2023). In the literature, victimization rates in studies reporting cyber violence range from approximately 5% to 90% (Cunha et al., 2024; Gilbar et al., 2023; Kim & Ferraresso, 2023; Melander & Hughes, 2018). Studies on college students estimate that the prevalence of dating violence, which includes physical, psychological, and sexual violence, ranges from 10% to 50%. In one study, approximately 50% of university students reported being victimized or perpetrated against by an intimate partner using communication technology (Henry et al., 2020). In another study conducted on individuals aged 18 and over, 14.85% of individuals reported having problems with at least one dimension of cyber aggression, cyber sexual coercion, cyber financial control, cyber control, and cyber stalking (Kim & Ferraresso, 2023). In another study, more than 20% of individuals reported that they exhibited behaviors such as impersonation and stalking toward their partners (López-Cepero et al., 2021). Our study revealed a low DIPV level among the students. However, the fact that the minimum and maximum scores were the same as the applied scale limits indicates that some individuals were exposed to or perpetrated high levels of digital violence. Therefore, dating violence among university students is a worrying issue.

Factors associated with DIPV in university students have been generally examined in three areas in the literature. First, sociodemographic factors such as age, gender, and community characteristics; second, developmental characteristics including family risk factors, history of violence, peer-related factors, psychological and behavioral factors, and cognition; and third, relational characteristics including relationship discord, conflict, and interaction patterns (Kim & Ferraresso, 2023; Kim et al., 2025). When the results of factors associated with DIPV were examined, younger age, lower education, lower self-esteem, poverty and socioeconomic disadvantage, more frequent drunkenness, substance use, sexual abuse, psychological distress such as attention deficit hyperactivity disorder, anxiety, stress, depression, mood disorder, higher levels of aggression, anxious or dismissive attachment styles, commitment to romantic love and justification of intimate partner violence, spending

more time online, and cyberaggression victimization were found to be associated with the following (Ahmadabadi et al., 2020; Cunha et al., 2022; Eriksson & Mazerolle, 2015; Jewkes et al., 2017; Kim & Ferraresso, 2023; Lelaurain et al., 2021; Maftai & Dănilă, 2023; Melander & Hughes, 2018; Wymbs et al., 2017). Our study indicates that digital dating violence scores have the strongest positive relationship with exposure to violence, perpetration of violence, and exposure to domestic violence. It is important to note that although these correlations reached statistical significance, their effect sizes were small ($r < .15$). Therefore, these associations should be interpreted with caution, as they do not reflect strong practical effects but rather suggest subtle patterns that may warrant further exploration in larger or more diverse samples. It is a striking finding that participants' exposure to domestic violence increases the DIPV score estimate of the model. This finding, in line with the literature, indicates that individuals who experience domestic violence may be more likely to experience violence from their partners in the digital environment. It has been reported that individuals who are directly exposed to, observe, and experience violence are more than 4 times more likely to perpetrate violence compared to individuals who report not being exposed to violence in childhood (Eriksson, & Mazerolle, 2015; UNFPA Türkiye, 2021). Domestic violence can cause individuals to internalize violence as a relationship norm and increase the risk of being exposed to partner violence later in life (Donat Bacioğlu, 2023). Young adults who experience physical abuse are thought to be more likely to accept aggression when it occurs, as such behaviors are an acceptable way to interact with loved ones (Melander & Hughes, 2018; Wymbs et al., 2017), which aligns with findings supporting the intergenerational transmission of violence.

Individuals report experiencing higher rates of victimization at younger ages (Grimani, et al., 2022). In our study, the digital violence score decreases as age increases. In addition, the fact that the DIPV score increases with the increase in the age variable shows that digital partner violence is not only common among young people. Although previous studies suggest that digital violence is more common in adolescents and young adults (Donat Bacioğlu, 2023), the findings of this study indicate that it does not decrease as age increases. In particular, the increase in seriousness of relationships and the strengthening of dependency dynamics in relationships may lead to an increase in digital control mechanisms between partners (UN Women, 2020). In another study conducted with participants under the age of 30, it was found that the probability of reporting victimization experiences was three times higher than for participants over the age of 30 (Henry et al., 2020). This finding may suggest that as participants age, the digital partner violence score

may increase, indicating that relationships could become more controlling or pressuring in the digital environment at older ages.

The fact that the participants' fathers were primary school graduates emerged as one of the factors that increased the model's violence estimate (Figure 5). It is known that low education levels reinforce traditional beliefs about gender roles and limit individuals' ability to establish egalitarian relationships (UN Women, 2020). Individuals raised by parents with low education levels may tend to normalize violence and experience a power imbalance in relationships. This result supports the literature emphasizing the direct relationship between gender equality and education levels. In our study, family income level and the individual's own income were found to be negatively associated with digital partner violence. In addition, the fact that the participant's family's income was equal to his expenses was considered one of the factors reducing the risk of digital partner violence. Achieving economic balance can enable the individual to be more independent in the relationship and reduce stress factors arising from economic difficulties. Financial security enables individuals to reject controlling or manipulative partner behaviors more easily (UNFPA Türkiye, 2021). Similarly, the individual's feeling that their own economic situation is in balance can positively affect the power balance in the relationship and reduce the risk of violence. Economic stability may contribute to individuals being less likely to engage in violent dynamics in their relationships. Poverty and socio-economic disadvantage place demands on close relationships and provide fertile ground for disagreements and conflict. In low-income families, both partners experience higher levels of violence. Studies report higher rates of violence in relationships where the husband earns a low income (Ahmadabadi et al., 2020). As a result, economic hardship creates a context that facilitates violence for both partners in a relationship.

In our study, the increase in digital partner violence scores among participants in metropolitan cities is linked to the intensive use of digital communication tools. Metropolitan cities with high internet and social media usage rates can heighten the risk of individuals encountering digital violence (UN Women, 2020). Indeed, the individualization and lack of privacy brought about by urbanization can increase controlling and oppressive digital behaviors between partners. In particular, situations such as individuals living in metropolitan cities being constantly followed by their partners and their accounts being monitored and checked in the digital environment can be evaluated within the scope of digital partner violence. These findings align with CRAT, suggesting that students' digital routines—such as extensive online communication and social media use—may create vulnerabilities by offering more opportunities for DIPV when capable guardianship is absent in cyberspace (Vakhitova, 2024). This theoretical integration grounds our

machine learning findings in a broader sociological and criminological framework, consistent with prior empirical work applying CRAT principles (Choi & Lee, 2017; Van Ouytsel et al., 2018).

The lower estimated digital partner violence score of participants from the Southeastern Anatolia Region (Figure 5) can be explained by regional cultural norms and social relations. In traditional social structures, the limited role of digital communication in partner relationships may be a factor in reducing digital partner violence. However, it should be kept in mind that this finding may also be related to regional awareness levels and reporting trends of digital violence (UNFPA Türkiye, 2021).

In our study, alcohol use and cigarette use also show a positive relationship with DIPV. This situation suggests that individuals who abstain from risky behaviors like alcohol and cigarettes can sustain healthier relationships. Individuals are more likely to be aggressive due to feelings of social disinhibition (Melander & Hughes, 2018). Cigarette use can be associated with high stress levels and risky behaviors. Smokers often experience increased anxiety, depression, and impulsivity, which can lead to relationship violence.

In our study, it is seen that there are significant inverse relationships between the duration of the relationship and the presence of a date and DIPV. Previous studies have indicated that digital partner violence is generally more common in long-term relationships (Donat Bacioğlu, 2023). The model evaluated the participant's low communication with his/her partner in the digital environment as increasing the risk. This situation suggests that low communication can lead to trust issues or control behaviors by the partner. The duration of the relationship (the participant's relationship lasting less than 1 year) had a decreasing effect on the digital violence score. The newness of the relationship may indicate that violent behaviors such as control and oppression have not yet developed.

The model considered the participant's limited time communicating with their partner in a digital environment as an increase in risk. Limited digital communication can lead to distrust and lack of communication between partners, which can increase the risk of digital violence. There is evidence in the literature that healthy communication between couples reduces digital partner violence (UN Women, 2020).

These results also resonate with findings from Türkiye. Prior studies among Turkish university students revealed that motivations such as anger, jealousy, and fear of loss frequently underpin digital dating violence behaviors (Bacioğlu & Akçay, 2024). Surveys among Turkish nursing students similarly indicated awareness and experiences of cyber-dating abuse within healthcare-related cohorts, underlining the importance of digital

literacy and prevention (Er et al., 2022). Furthermore, attachment styles were found to be predictive of cyber-dating violence behaviors in Turkish samples (Erdem et al., 2022). More recently, mixed-methods research has highlighted the persistence of psychological violence in intimate partner relationships (Keçeci & Ümmet, 2025), while regional studies report notable prevalence of dating violence in university populations in Eastern Türkiye (Calikoğlu et al., 2021). Together, these findings confirm that cultural and contextual factors in Türkiye play a critical role in shaping the dynamics of DIPV.

The results of the study are consistent with previous findings that tree-based algorithms such as XGBoost and RF are successful in violence prediction (Low et al., 2024). It has also been reported in the literature that deep tree-based models are more successful than other ML methods in predicting situations associated with complex, multivariate, and difficult-to-predict behaviors such as digital partner violence (Parmigiani et al., 2022). On the other hand, the low performance of SVM is also parallel to studies showing that kernel-based methods may have difficulty capturing nonlinear relationships, especially in large-scale datasets (Hui et al., 2023). Our findings align with prior applied ML studies in the field of violence research, which also underscore the potential of ML to reveal hidden dependencies and improve predictive accuracy (Cruz-Mendoza et al., 2025; Erbiçer et al., 2025; Hui et al., 2023). In addition, the high accuracy of RF and GB models indicates the potential of these models to be integrated into clinical decision support systems in identifying individuals at risk of digital violence (Low et al., 2024). While traditional statistical techniques remain important for hypothesis-driven analyses, the application of ML in this study demonstrates added value in terms of predictive power and flexibility. In particular, ML's ability to reveal hidden interactions between demographic, psychosocial, and behavioral factors provides novel insights into DIPV that might not emerge from conventional regression or structural models. Moreover, our results echo findings from these applied ML studies in IPV research (Chen et al., 2023; Shashidhara et al., 2024), which show ML's ability to unveil complex risk relationships. Coupled with SHAP's interpretability, our approach provides not only predictive power but also theoretically meaningful insights by highlighting contributions of digital communication and prior exposure—factors consistent with theoretical models such as CRAT. Beyond statistical performance, the identification of factors such as prolonged digital communication and prior exposure to domestic violence as important predictors resonates with CRAT. These findings suggest that routine online behaviors may increase exposure and opportunity for DIPV when guardianship mechanisms are limited, consistent with both criminological theory and prior IPV literature.

This study demonstrates that ML models are effective in predicting the risk of digital partner violence, and XGBoost in particular stands out as the model with the highest discrimination power. However, it should be emphasized that due to the cross-sectional nature of this study, new studies with longitudinal datasets are needed to evaluate the performance of the models over time. At the same time, the very high AUC values observed may partly reflect overfitting. Although stratified cross-validation reduced this risk, it cannot be fully eliminated, and the generalizability of the models should therefore be interpreted with caution.

From a policy and practice perspective, these findings suggest several targeted directions. First, digital literacy programs in universities should not only focus on general awareness but specifically address overcommunication, surveillance behaviors, and online control mechanisms identified as risk factors in this study. Second, students with a history of family or domestic violence may benefit from early counseling and support services, as their elevated risk of DIPV was evident in our findings. Third, socioeconomic disadvantage was shown to exacerbate vulnerability, suggesting the need for structural support and financial counseling initiatives. Finally, the strong predictive performance of tree-based ML models indicates the potential to develop screening tools that could be integrated into student counseling or university health services to identify at-risk individuals in a timely manner.

Finally, in line with the *Journal of Interpersonal Violence's* policy on diversity, it is important to situate our findings within broader issues of representation and generalizability. The current study was conducted with students from a single public university in Türkiye, the majority of whom were female. While this reflects the gender distribution in some higher education contexts, it also limits the ability to generalize findings across male students, non-binary individuals, or those from different educational or cultural backgrounds. Moreover, regional variations observed in our study highlight how cultural context shapes DIPV dynamics, suggesting the importance of comparative research across diverse settings. Future research should therefore engage with more heterogeneous samples, including students from different geographic regions, socioeconomic statuses, and gender/sexual identity groups, to ensure that diversity in experiences of DIPV is better captured and addressed.

Limitations

Since the study has a cross-sectional design, it is not possible to establish a cause–effect relationship. Longitudinal studies are needed to understand the changes in digital partner violence over time. The sample of the study is limited to students from a single public university in Türkiye. In addition, since

the snowball sampling method was used, the individuals participating in the study may have been concentrated in certain social circles. Furthermore, as the study sample consisted only of students from a single public university in Türkiye, the generalizability of the findings to other cultural and educational contexts is limited. Future studies should include larger and more diverse samples from multiple universities and countries to enhance external validity.

Since the data collection method used in the study is self-report questionnaires, the responses given by the participants may vary due to cognitive biases, false recall, or social desirability effects. The DIPVS used in the study measures only certain aspects of digital violence. Broad dimensions such as emotional, psychological, or financial digital violence were not directly addressed. This may cause some types of digital violence to not be fully detected. The use of more comprehensive scales and data collection tools that include different psychosocial factors is an important area of development for future research. In this study, only six different ML models (LR, XGBoost, Gradient Boosting, Random Forest, LightGBM, SVM) were used. In future studies, more comprehensive analyses can be performed with deep learning (DL) methods (e.g., convolutional neural network (CNN), recurrent neural network (RNN), or transformer-based models). Furthermore, as the study sample consisted only of Turkish university students from a single institution, the generalizability of the findings to other cultural and educational contexts is limited. Future studies should include larger and more diverse samples from multiple universities and countries to enhance external validity. Although stratification was applied to maintain proportionality, the dataset still reflected an imbalance between positive and negative cases. Future studies may consider applying advanced resampling techniques (e.g., SMOTE) or class-weighting approaches to further improve model robustness.

Generalizability

Since the study covers students from a specific university, its generalizability is limited to student groups with similar demographic structures. However, it is possible to make inferences for individuals with similar age groups in other universities in Turkey. Risk factors associated with digital partner violence show similar trends in the literature with studies conducted in different countries. Particularly, there is universal acceptance of the effects of variables such as domestic violence, socioeconomic status, education level, and psychological factors on digital violence. In this respect, the findings may also be valid in different cultural contexts. The ML models used can be adapted to different data sets, allowing similar analyses to be conducted on other groups. For example, similar prediction models can be used when studying different

age groups (middle-aged, elderly individuals). These models can be adapted for violence types other than digital partner violence (e.g., cyberbullying or domestic digital violence). Furthermore, generalizability is limited by the reliance on a single university sample, the predominance of female participants, and the use of snowball sampling. These factors may constrain the extent to which the findings apply to broader or more diverse populations.

Conclusion

This study has demonstrated that ML models can make successful predictions by revealing the relationship between DIPV and individual, socioeconomic, and relational factors. The findings showed that digital partner violence is related to individual, familial, and relational factors. Specifically, it was found that the DIPV score goes up when a person has experienced domestic violence, is older, lives in a big city, and communicates with their partner for a short time online. On the other hand, the risk of digital partner violence goes down when a person has been in a relationship for a long time, thinks their family's income is stable, and doesn't smoke.

When evaluated in terms of ML models, the XGBoost model (99.6% AUC-ROC) stood out as the model with the best discrimination power. RF (99.5% AUC-ROC) and LightGBM (99.5% AUC-ROC) models also showed strong performance. SVM showed significantly lower accuracy and F1 score compared to other models. These results reveal that tree-based ML algorithms are more successful in predicting digital partner violence.

The study offers important implications in terms of preventing digital partner violence and raising awareness. However, given the limitations, we need more comprehensive studies to apply these results to diverse populations. In line with the results, it can be suggested that digital literacy and cybersecurity training be disseminated at the individual level, access to psychosocial support services be increased, and awareness-raising studies be conducted at universities. In the academic field, longitudinal studies can be used to examine the change in digital violence over time, cultural differences can be analyzed with larger samples, and modeling processes can be developed with DL techniques. On the legal level, digital partner violence can be defined more clearly within the legal framework, criminal sanctions can be increased, and algorithms can be developed to prevent violence on social media platforms. Media campaigns can be organized for social awareness, digital security topics can be added to educational curricula, and programs can be implemented to guide young individuals in establishing healthy relationships. To prevent digital partner violence, it can be suggested that

individual awareness be increased, legal regulations be strengthened, academic research be expanded, and technological solutions be implemented.

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Ethical Considerations

Permission was obtained from the Çanakkale Onsekiz Mart University Graduate Education Institute Ethics Committee for the conduct of the study (Decision Number: 2024-YÖNP-0226/05/79).

Consent to Participate

Written informed consent was also obtained from the participants. There was an informed consent form at the beginning of the questionnaire, and there was a check box in the questionnaire indicating that it was completed voluntarily. Participants were informed that participation in the study was voluntary and that the data collected would only be used for scientific purposes.

Author Contributions

GÖ: Study conception and design, Data collection, Data analysis and interpretation, Drafting of the article, Critical revision of the article.

HK: Study conception and design, Data collection, Drafting of the article, Critical revision of the article.

HY: Study conception and design, Data collection.

AO: Study conception and design, Data collection.

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